# Clustering

Readings :
Chapters 11, 12, 13, 14 “Pattern recognition” Theodoridis & Koutroumbas

Chapter 9 “Pattern recognition and machine learning” Bishop

# Clustering

**What is a cluster?**

Identifying groups of instances such that the similarities between an instance and other instances in the same group are larger than the similarities with instances from other groups.

**How can a cluster be built?**

- Expert knowledge
- Automatic induction from data $D \equiv \{\mathbf{x}_n\}_{n=1}^{N}$

$S(N,K)$: # of different possible assignments of N points to K clusters

$$S(N,1) = S(N,N) = 1$$

$$S(N,K) = K\,S(N-1,K) + S(N-1,K-1)$$

either the Nth point is assigned to one of K existing clusters or the Nth point forms a cluster by itself.

$$S(N,K) = \frac{1}{K!}\sum_{i=0}^{K}(-1)^{K-i}\binom{K}{i} i^N \quad \text{[Stirling numbers of the second kind]}$$

E.g. $S(100,5) \approx 10^{68}$

**Tasks for building clusters**

- Representation of instances as vectors of features
- Feature selection
- Definition of similarity or dissimilarity measures
- Choice of clustering criterion
- Clustering procedure
- Validation of results
- Interpretation of results

# Proximity between points

Given $\mathbf{x}, \mathbf{x}' \in X$

**Consider the map** $d: X \times X \rightarrow R$

$\mathbf{x}, \mathbf{x}'$ $\quad$ $d(\mathbf{x},\mathbf{x}')$

**The map** $d(\mathbf{x},\mathbf{x}')$ **is a dissimilarity measure if**

(i) $0 \leq d(\mathbf{x},\mathbf{x}') < \infty$
(ii) $d(\mathbf{x},\mathbf{x}) = 0$
(iii) $d(\mathbf{x},\mathbf{x}') = d(\mathbf{x}',\mathbf{x})$ [Symmetric]

**The map** $d(\mathbf{x},\mathbf{x}')$ **is a metric dissimilarity measure if**

(iv) $d(\mathbf{x},\mathbf{x}') = 0 \Leftrightarrow \mathbf{x} = \mathbf{x}'$
(v) $d(\mathbf{x},\mathbf{x}'') \leq d(\mathbf{x},\mathbf{x}') + d(\mathbf{x}',\mathbf{x}'')$

E.g. The Minkowski metric $d_q(\mathbf{x},\mathbf{x}') = \left[\sum_{d=1}^{D} |x_d - x'_d|^q\right]^{1/q}$ ; $\mathbf{x},\mathbf{x}' \in \Re^D$

$q = 1$ $\quad$ $d_1(\mathbf{x},\mathbf{x}') = \sum_{d=1}^{D} |x_d - x'_d|$ $\quad$ [Manhattan distance]

$q = 2$ $\quad$ $d_2(\mathbf{x},\mathbf{x}') = \sqrt{\sum_{d=1}^{D} (x_d - x'_d)^2}$ $\quad$ [Euclidean distance]

$q = \infty$ $\quad$ $d_\infty(\mathbf{x},\mathbf{x}') = \max_{d=1,\ldots,D} |x_d - x'_d|$ $\quad$ [$L_\infty$ distance]

# Proximity between a point and a cluster

Given the instance **x** and the cluster *C* (size *N*) it is possible to define different measures of proximity between a point and a set

1. **The min function**

$$d_{\min}(\mathbf{x}, C) = \min_{\mathbf{x}' \in C} d(\mathbf{x}, \mathbf{x}')$$

2. **The max function**

$$d_{\max}(\mathbf{x}, C) = \max_{\mathbf{x}' \in C} d(\mathbf{x}, \mathbf{x}')$$

3. **The mean function**

$$d_{\text{mean}}(\mathbf{x}, C) = \frac{1}{N} \sum_{\mathbf{x}' \in C} d(\mathbf{x}, \mathbf{x}')$$

- Other possibilities
  Characterize the cluster by a **representative point** (e.g. the center of mass)
  **distance between an instance and a cluster** = **distance between the instance and the representative point of the cluster**.

# Proximity between clusters

Given the clusters $C_1$ (size $N_1$) and $C_2$ (size $N_2$), it is possible to define different measures of proximity for sets

1. **The min function (single link)**

$$d_{\min}(C_1, C_2) = \min_{\mathbf{x}_1 \in C_1, \mathbf{x}_2 \in C_2} d(\mathbf{x}_1, \mathbf{x}_2)$$

2. **The max function (complete link)**

$$d_{\max}(C_1, C_2) = \max_{\mathbf{x}_1 \in C_1, \mathbf{x}_2 \in C_2} d(\mathbf{x}_1, \mathbf{x}_2)$$

3. **The mean function**

$$d_{\text{mean}}(C_1, C_2) = \frac{1}{N_1 N_2} \sum_{\mathbf{x}_1 \in C_1, \mathbf{x}_2 \in C_2} d(\mathbf{x}_1, \mathbf{x}_2)$$

**Note: Even if** d **is a dissimilarity measure**
$d_{min}$ , $d_{max}$ , $d_{mean}$ **are not measures**

- Other possibilities
  Characterize the clusters by a **representative point** (e.g. the center of mass)
  **distance between clusters** =
  **distance between representative points**.

# Agglomerative hierarchical clustering

**Name:** **Agglomerative hierarchical clustering**

**Input:** Instances: $\{\mathbf{x}_n; n = 1,2,\ldots,N\}$

Matrix of distances among instances: $D_{nm} = d(\mathbf{x}_n, \mathbf{x}_m)$

distance function: $g(d, d')$

[min, max, mean, weighted average, etc.]

**Output:** Dendrogram

**Code:** Initialize clusters $R_0 = \{\{x_n\}; n = 1,2,\ldots,N\}$

$K = N$

$n = 0$

while $K > 1$

$\mathrm{n} = \mathrm{n} + 1$

Find $C_i, C_j$ such that $d(C_i, C_j) = \min_{r,s=1,\ldots,K} d(C_r, C_s)$

$\mathrm{R_n} = (\mathrm{R_{n-1}} - \{C_i, C_j\}) \cup \{C_i \cup C_j\}$

$K = K - 1$

for $k = 1:K, \quad k \neq i, k \neq j$

calculate $d(C_i \cup C_j, C_k) = g(d(C_i, C_k), d(C_j, C_k))$

Relabel clusters so that $R_i = \{C_i; \quad i = 1,2,\ldots,K\}$

end

# Agglomerative hierarchical clustering (single link)

distance function : $g(d,d') = \min(d, d')$

$$D(R_0) = \begin{array}{c} \\ \{1\} \\ \{2\} \\ \{3\} \\ \{4\} \\ \{5\} \end{array} \begin{array}{c} \begin{array}{ccccc} \{1\} & \{2\} & \{3\} & \{4\} & \{5\} \end{array} \\ \begin{pmatrix} 0 & & & & \\ 9 & 0 & & & \\ 3 & 7 & 0 & & \\ 6 & 5 & 9 & 0 & \\ 11 & 10 & \mathbf{2} & 8 & 0 \end{pmatrix} \end{array}$$

$$D(R_1) = \begin{array}{c} \\ \{3,5\} \\ \{1\} \\ \{2\} \\ \{4\} \end{array} \begin{array}{c} \begin{array}{cccc} \{3,5\} & \{1\} & \{2\} & \{4\} \end{array} \\ \begin{pmatrix} 0 & & & \\ \mathbf{3} & 0 & & \\ 7 & 9 & 0 & \\ 8 & 6 & 5 & 0 \end{pmatrix} \end{array}$$

$$D(R_2) = \begin{array}{c} \\ \{1,3,5\} \\ \{2\} \\ \{4\} \end{array} \begin{array}{c} \begin{array}{ccc} \{1,3,5\} & \{2\} & \{4\} \end{array} \\ \begin{pmatrix} 0 & & \\ 7 & 0 & \\ 6 & \mathbf{5} & 0 \end{pmatrix} \end{array}$$

$$D(R_3) = \begin{array}{c} \\ \{1,3,5\} \\ \{2,4\} \end{array} \begin{array}{c} \begin{array}{cc} \{1,3,5\} & \{2,4\} \end{array} \\ \begin{pmatrix} 0 & \\ \mathbf{6} & 0 \end{pmatrix} \end{array}$$

$$D(R_4) = \{1,2,3,4,5\}$$

# Agglomerative hierarchical clustering (complete link)

distance function : $g(d,d') = \max(d, d')$

$$D(R_0) = \begin{array}{c} \\ \{1\} \\ \{2\} \\ \{3\} \\ \{4\} \\ \{5\} \end{array} \begin{array}{c} \begin{array}{ccccc} \{1\} & \{2\} & \{3\} & \{4\} & \{5\} \end{array} \\ \begin{pmatrix} 0 & & & & \\ 9 & 0 & & & \\ 3 & 7 & 0 & & \\ 6 & 5 & 9 & 0 & \\ 11 & 10 & \mathbf{2} & 8 & 0 \end{pmatrix} \end{array}$$

$$D(R_1) = \begin{array}{c} \\ \{3,5\} \\ \{1\} \\ \{2\} \\ \{4\} \end{array} \begin{array}{c} \begin{array}{cccc} \{3,5\} & \{1\} & \{2\} & \{4\} \end{array} \\ \begin{pmatrix} 0 & & & \\ 11 & 0 & & \\ 10 & 9 & 0 & \\ 9 & 6 & \mathbf{5} & 0 \end{pmatrix} \end{array}$$

$$D(R_2) = \begin{array}{c} \\ \{3,5\} \\ \{2,4\} \\ \{1\} \end{array} \begin{array}{c} \begin{array}{ccc} \{3,5\} & \{2,4\} & \{1\} \end{array} \\ \begin{pmatrix} 0 & & \\ 10 & 0 & \\ 11 & \mathbf{9} & 0 \end{pmatrix} \end{array}$$

$$D(R_3) = \begin{array}{c} \\ \{3,5\} \\ \{1,2,4\} \end{array} \begin{array}{c} \begin{array}{cc} \{3,5\} & \{1,2,4\} \end{array} \\ \begin{pmatrix} 0 & \\ \mathbf{11} & 0 \end{pmatrix} \end{array}$$

$$D(R_4) = \{1,2,3,4,5\}$$

# Agglomerative hierarchical clustering (average link)

distance function : $g(d,d') = \text{mean}(d, d')$

$$D(R_0) = \begin{array}{c} \\ \{1\} \\ \{2\} \\ \{3\} \\ \{4\} \\ \{5\} \end{array} \begin{array}{c} \begin{array}{ccccc} \{1\} & \{2\} & \{3\} & \{4\} & \{5\} \end{array} \\ \begin{pmatrix} 0 & & & & \\ 9 & 0 & & & \\ 3 & 7 & 0 & & \\ 6 & 5 & 9 & 0 & \\ 11 & 10 & \mathbf{2} & 8 & 0 \end{pmatrix} \end{array}$$

$$D(R_1) = \begin{array}{c} \\ \{3,5\} \\ \{1\} \\ \{2\} \\ \{4\} \end{array} \begin{array}{c} \begin{array}{cccc} \{3,5\} & \{1\} & \{2\} & \{4\} \end{array} \\ \begin{pmatrix} 0 & & & \\ 7 & 0 & & \\ 8.5 & 9 & 0 & \\ 8.5 & 6 & \mathbf{5} & 0 \end{pmatrix} \end{array}$$

$$D(R_2) = \begin{array}{c} \\ \{3,5\} \\ \{2,4\} \\ \{1\} \end{array} \begin{array}{c} \begin{array}{ccc} \{3,5\} & \{2,4\} & \{1\} \end{array} \\ \begin{pmatrix} 0 & & \\ 8.5 & 0 & \\ \mathbf{7} & 7.5 & 0 \end{pmatrix} \end{array}$$

$$D(R_3) = \begin{array}{c} \\ \{1,3,5\} \\ \{2,4\} \end{array} \begin{array}{c} \begin{array}{cc} \{1,3,5\} & \{2,4\} \end{array} \\ \begin{pmatrix} 0 & \\ \mathbf{8} & 0 \end{pmatrix} \end{array}$$

$$D(R_4) = \{1,2,3,4,5\}$$

# Agglomerative sequential clustering (I)

New instances are either assigned to an existing cluster or forms a new cluster, depending on the distances to the existing clusters.

**Name:** **Agglomerative sequential clustering**

**Input:** Instances: $\{\mathbf{x}_n; n = 1,2,\ldots,N\}$

Distance function between point and cluster: $d(\mathbf{x}_n, C_i)$

[min, max, mean, etc.]

Threshold: $\theta$

**Output:** Partition of instances into clusters

**Code:** Initialize cluster $C_1 = \{\mathbf{x}_1\}$

$K = 1$

for n = 2 :N

Find $C_i$ such that $d(\mathbf{x}_n, C_i) = \min_{j=1,\ldots,K} d(\mathbf{x}_n, C_j)$

if $d(\mathbf{x}_n, C_i) \leq \theta$

$C_i = C_i \cup \{\mathbf{x}_n\}$

else

$K = K + 1$

$C_K = \{\mathbf{x}_n\}$

end

end

# Agglomerative sequential clustering (II)

- The number and configuration of the clusters obtained depend on the value of the threshold and on the order of presentation of the instances.

- The number of clusters could be determined using the following algorithm

**Name:** **Determination of number of clusters/threshold value**

**Input:** Instances: $\{\mathbf{x}_n; n = 1,2,\ldots,N\}$

$$a = \min_{i,j=1,2,\ldots,N;\ i\neq j} d(\mathbf{x}_i, \mathbf{x}_j)$$

$$b = \max_{i,j=1,2,\ldots,N;\ i\neq j} d(\mathbf{x}_i, \mathbf{x}_j)$$

$c$ : Step size for grid of threshold values

$M$ : Number of simulations

**Output:** Threshold $\theta^*$

**Code:** Initialize cluster $C_1 = \{\mathbf{x}_1\}$

$K = 1$

for $\theta = a$ to $b$ in steps of $c$

- Run sequential clustering M times with parameter $\theta$ and random order in the presentation of instances.
- Estimate the number of clusters $K(\theta)$ as the most frequent size obtained in the $M$ executions.

end

Estimate the optimal values of $\theta^*, K(\theta^*)$ from a plot $K(\theta^*)$ vs. $\theta^*$

# Agglomerative sequential clustering (III)

Choose the value of the threshold $\theta^*$ in one of the larger flat regions (not necessarily the largest)

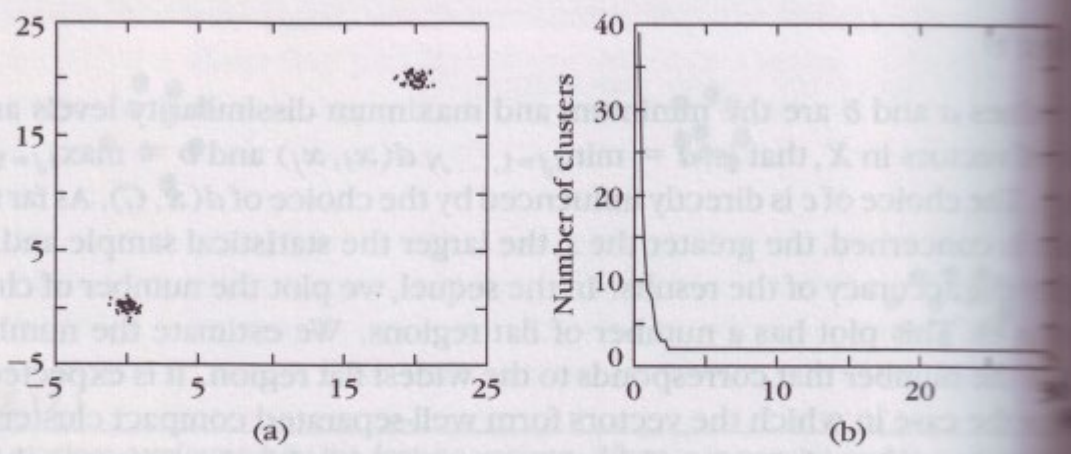



**FIGURE 12.2**

(a) The data set. (b) The plot of the number of clusters versus $\Theta$. It can be seen that the range of values of $\Theta$, the number of clusters, $m$, is 2.

Source: “Pattern recognition” Theodoridis & Koutroumbas

# Agglomerative sequential clustering (example)

Distance function : $d(\mathbf{x}_i, C_k) = \frac{1}{N_k} \sum_{x_j \in C_k} d(\mathbf{x}_i, \mathbf{x}_j)$

Threshold : $\theta = 6$

$$d_{ij} \equiv d(\mathbf{x}_i, \mathbf{x}_j) = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{pmatrix} 0 & & & & \\ 9 & 0 & & & \\ 3 & 7 & 0 & & \\ 6 & 5 & 9 & 0 & \\ 11 & 10 & 2 & 8 & 0 \end{pmatrix} \end{array}$$

$\mathbf{x}_1$: $C_1 = \{\mathbf{x}_1\}$

$\mathbf{x}_2$: $d(\mathbf{x}_2, C_1) = 9$ (MIN $> \theta$)

$C_2 = \{\mathbf{x}_2\}$

$\mathbf{x}_3$: $d(\mathbf{x}_3, C_1) = 3$ (MIN); $d(\mathbf{x}_3, C_2) = 7$;

$C_1 = \{\mathbf{x}_1, \mathbf{x}_3\}$

$\mathbf{x}_4$: $d(\mathbf{x}_4, C_1) = 7.5$; $d(\mathbf{x}_4, C_2) = 5$ (MIN);

$C_2 = \{\mathbf{x}_2, \mathbf{x}_4\}$

$\mathbf{x}_5$: $d(\mathbf{x}_5, C_1) = 6.5$ (MIN $> \theta$); $d(\mathbf{x}_5, C_2) = 9$;

$C_3 = \{\mathbf{x}_5\}$

$OUTPUT : \{\{\mathbf{x}_1, \mathbf{x}_3\}, \{\mathbf{x}_2, \mathbf{x}_4\}, \{\mathbf{x}_5\}\}$

# Clustering languages

TABLE 12.3 NUMERALS IN 11 LANGUAGES

| English (E) | Norwegian (N) | Danish (Da) | Dutch (Du) | German (G) | French (Fr) | Spanish (Sp) | Italian (I) | Polish (P) | Hungarian (H) | Finnish (Fi) |
|---|---|---|---|---|---|---|---|---|---|---|
| one | en | en | een | eins | un | uno | uno | jeden | egy | yksi |
| two | to | to | twee | zwei | deux | dos | due | dwa | ketto | kaksi |
| three | tre | tre | drie | drei | trois | tres | tre | trzy | harom | kolme |
| four | fire | fire | vier | vier | quatre | cuatro | quattro | cztery | negy | neua |
| five | fem | fem | vijf | funf | cinq | cinco | cinque | piec | ot | viisi |
| six | seks | seks | zes | sechs | six | seis | sei | szesc | hat | kuusi |
| seven | sju | syv | zeven | sieben | sept | siete | sette | siedem | het | seitseman |
| eight | atte | otte | acht | acht | huit | ocho | otto | osiem | nyolc | kahdeksan |
| nine | ni | ni | negen | neun | neuf | nueve | nove | dziewiec | kilenc | yhdeksan |
| ten | ti | ti | tien | zehn | dix | diez | dieci | dziesiec | tiz | kymmenen |

Concordance of first letters= ...

| | En | No | Da | Du | Ge | Fr | Sp | It | Pol | Hu | Fi |
|---|---|---|---|---|---|---|---|---|---|---|---|
| En | 10 | 8 | 8 | 3 | 4 | 4 | 4 | 4 | 3 | 1 | 1 |
| No | 8 | 10 | 9 | 5 | 6 | 4 | 4 | 4 | 3 | 2 | 1 |
| Da | 8 | 9 | 10 | 4 | 5 | 4 | 5 | 5 | 4 | 2 | 1 |
| Du | 3 | 5 | 4 | 10 | 5 | 1 | 1 | 1 | 0 | 2 | 1 |
| Ge | 4 | 6 | 5 | 5 | 10 | 3 | 3 | 3 | 2 | 1 | 1 |
| Fr | 4 | 4 | 4 | 1 | 3 | 10 | 8 | 9 | 5 | 0 | 1 |
| Sp | 4 | 4 | 4 | 1 | 3 | 8 | 10 | 9 | 7 | 0 | 1 |
| It | 4 | 4 | 5 | 1 | 3 | 9 | 9 | 10 | 6 | 0 | 1 |
| Pol | 3 | 3 | 4 | 0 | 2 | 5 | 7 | 6 | 10 | 0 | 1 |
| Hu | 1 | 2 | 2 | 2 | 1 | 0 | 0 | 0 | 0 | 10 | 2 |
| Hu | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 10 |

# Sequential clustering for languages

Distance function : $d(\mathbf{x}_i, C_k) = \frac{1}{N_k} \sum_{x_j \in C_k} d(\mathbf{x}_i, \mathbf{x}_j)$

Threshold : $\theta = 5$

$$d_{ij} \equiv d(\mathbf{x}_i, \mathbf{x}_j) = \begin{array}{c} \\ En \\ It \\ Pol \\ Da \\ Ge \end{array} \begin{array}{c} \begin{array}{ccccc} En & It & Pol & Da & Ge \end{array} \\ \begin{pmatrix} 0 & & & & \\ 1 & 0 & & & \\ 3 & 4 & 0 & & \\ 5 & 5 & 6 & 0 & \\ 7 & 7 & 8 & 5 & 0 \end{pmatrix} \end{array}$$

$En$: $C_1 = \{En\}$

$It$: $d(It, C_1) = 1$ (MIN $\leq \theta$)

$C_1 = \{En, It\}$

$Pol$: $d(Pol, C_1) = 3.5$ (MIN $\leq \theta$);

$C_1 = \{En, It, Pol\}$

$Da$: $d(Da, C_1) = 16/3$ (MIN $> \theta$);

$C_2 = \{Da\}$

$De$: $d(De, C_1) = 22/3$; $d(De, C_2) = 5$ (MIN $\leq \theta$);

$C_2 = \{Da, De\}$

$OUTPUT : \{\{En, It, Pol\}, \{Da, De\}\}$

# K-means

Assign $N$ instances $\{\mathbf{x}_n\}_{n=1}^N$ to $K$ clusters, where each of the clusters is characterized by representative vectors $\{\boldsymbol{\mu}_k\}_{k=1}^K$, so as to minimize the **distortion measure**

$$\mathrm{J} = \sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk}\left\|\mathbf{x}_n - \boldsymbol{\mu}_k\right\|^2 \qquad r_{nk} = \begin{cases} 1 & \text{if } \mathbf{x}_n \in C_k \\ 0 & \text{if } \mathbf{x}_n \notin C_k \end{cases}$$

**GOAL:** Find optimal values of

$$\{\boldsymbol{\mu}_k; \{r_{nk}; n = 1,\ldots N\}\}_{k=1}^K$$

- Assume $\{\boldsymbol{\mu}_k\}_{k=1}^K$ known, optimize $\{r_{nk}; n = 1,\ldots N\}_{k=1}^K$

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg\min_{j=1,2,\ldots,K}\left\|\mathbf{x}_n - \boldsymbol{\mu}_j\right\|^2 \\ 0 & \text{otherwise} \end{cases}$$

**Expectation step**

- Assume $\{r_{nk}; n = 1,\ldots N\}_{k=1}^K$ known, optimize $\{\boldsymbol{\mu}_k\}_{k=1}^K$

$$\frac{\partial \mathrm{J}}{\partial \boldsymbol{\mu}_k} = 2\sum_{n=1}^{N} r_{nk}\left(\mathbf{x}_n - \boldsymbol{\mu}_k\right);$$

**Maximization step**

$$\frac{\partial \mathrm{J}}{\partial \boldsymbol{\mu}_k} = 0 \quad \Rightarrow \quad \sum_{n=1}^{N} r_{nk}\left(\mathbf{x}_n - \boldsymbol{\mu}_k\right) = 0 \quad \Rightarrow \quad \boldsymbol{\mu}_k = \frac{1}{N_k}\sum_{n=1}^{N} r_{nk}\mathbf{x}_n;$$

$\boldsymbol{\mu}_k$ is the average of the $N_k$ instances assigned to cluster k

$$N_k = \sum_{n=1}^{N} r_{nk}$$

# K-means algorithm

**Name:** **K-means clustering (batch)**

**Input:** Instances: $\{\mathbf{x}_n; n = 1,2,\ldots,N\}$

$K$: Number of clusters

**Output:** Clusters $\{r_{nk;}\ k=1,\ldots,K; N=1,\ldots,N\}$ & cluster means $\{\mu_{k;}; k=1,\ldots,K\}$

**Code:** Initialize the values of the cluster means $\boldsymbol{\mu}_k$

E.g. $\boldsymbol{\mu}_k = \mathbf{x}_{n_k}; \quad k = 1,2,\ldots,K$

Repeat

1. **Expectation step**

for $n = 1:N$

$$i = \arg\min_{k=1,\ldots,K} \left|\mathbf{x}_n - \boldsymbol{\mu}_k\right|^2$$

$$r_{ni} = 1; \qquad r_{nk} = 0, \quad k = 1,2,\ldots,K \quad k \neq i$$

end

Hard (crisp) cluster assignment

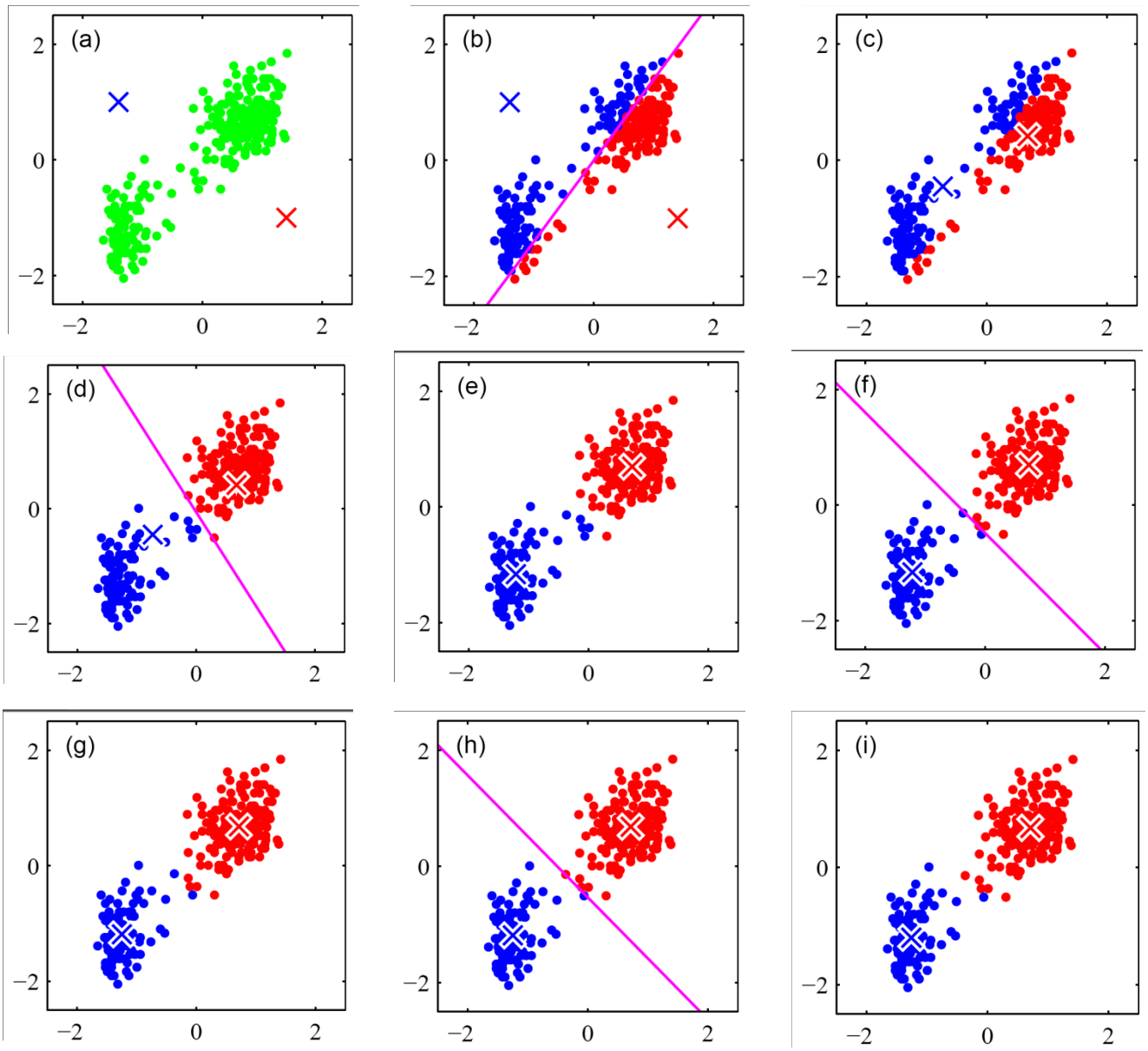
2. **Maximization step**

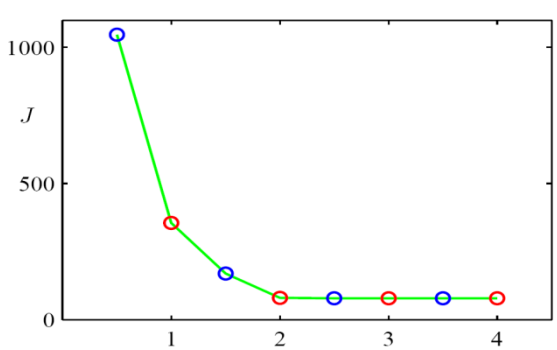
$$\boldsymbol{\mu}_k = \frac{1}{N_k}\sum_{n=1}^{N} r_{nk}\mathbf{x}_n;$$

$$N_k = \sum_{n=1}^{N} r_{nk}; \qquad k = 1,2,\ldots,K$$

until convergence (based on improvement of J)

# Fuzzy (soft) K-means



# Image segmentation and compression

**Figure 9.3** Two examples of the application of the $K$-means clustering algorithm to image segmentation showing the initial images together with their $K$-means segmentations obtained using various values of $K$. This also illustrates of the use of vector quantization for data compression, in which smaller values of $K$ give higher compression at the expense of poorer image quality.

# Fuzzy (soft) K-means

Assign *N* instances $\{\mathbf{x}_n\}_{n=1}^{N}$ to *K* clusters, where each of the clusters is characterized by representative vectors $\{\boldsymbol{\mu}_k\}_{k=1}^{K}$ , so as to minimize the **distortion measure**

$$\mathrm{J} = \sum_{n=1}^{N}\sum_{k=1}^{N} u_{nk}^2 \left\|\mathbf{x}_n - \boldsymbol{\mu}_k\right\|^2$$

$$u_{nk} \in [0,1] \quad [\text{membership of instance } n \text{ to cluser } k]$$

$$\sum_{\mathrm{k}=1}^{\mathrm{K}} u_{nk} = 1; \quad n = 1,2,\ldots,N \qquad [\text{normalization}]$$

**OPTIMIZATION**

$$\mathrm{J} = \sum_{n=1}^{N}\sum_{k=1}^{N} u_{nk}^2 \left\|\mathbf{x}_n - \boldsymbol{\mu}_k\right\|^2 - \sum_{n=1}^{N}\lambda_n\left(\sum_{k=1}^{N} u_{nk} - 1\right)$$

$$\frac{\partial J}{\partial u_{nk}} = 2u_{nk}\left\|\mathbf{x}_n - \boldsymbol{\mu}_k\right\|^2 - \lambda_n$$

$$\frac{\partial J}{\partial \boldsymbol{\mu}_k} = -2\sum_{n=1}^{N} u_{nk}^2\left(\mathbf{x}_n - \boldsymbol{\mu}_k\right)$$

# Optimization for fuzzy (soft) K-means

$$\mathrm{J} = \sum_{n=1}^{N}\sum_{k=1}^{N} u_{nk}^2 \left\| \mathbf{x}_n - \boldsymbol{\mu}_k \right\|^2 - \sum_{n=1}^{N} \lambda_n \left( \sum_{k=1}^{N} u_{nk} - 1 \right)$$

$$\frac{\partial J}{\partial u_{nk}} = 0 \quad \Rightarrow \quad 2u_{nk}\left\| \mathbf{x}_n - \boldsymbol{\mu}_k \right\|^2 - \lambda_n = 0 \quad \Rightarrow \quad u_{nk} = \frac{\lambda_n}{2\left\| \mathbf{x}_n - \boldsymbol{\mu}_k \right\|^2}$$

$$1 = \frac{1}{2}\lambda_n \sum_{j=1}^{K} \left\| \mathbf{x}_n - \boldsymbol{\mu}_j \right\|^{-2} \quad \Rightarrow \quad \lambda_n = \frac{2}{\sum_{k=1}^{K} \left\| \mathbf{x}_n - \boldsymbol{\mu}_j \right\|^{-2}}$$

$$u_{nk} = \frac{\left\| \mathbf{x}_n - \boldsymbol{\mu}_k \right\|^{-2}}{\sum_{k=1}^{K} \left\| \mathbf{x}_n - \boldsymbol{\mu}_j \right\|^{-2}}$$

$$\frac{\partial J}{\partial \boldsymbol{\mu}_k} = 0 \quad \Rightarrow \quad -2\sum_{n=1}^{N} u_{nk}^2 \left( \mathbf{x}_n - \boldsymbol{\mu}_k \right) = 0 \quad \Rightarrow$$

$$\boldsymbol{\mu}_k = \frac{1}{N_k}\sum_{n=1}^{N} u_{nk}^2 \mathbf{x}_n; \qquad N_k = \sum_{n=1}^{N} u_{nk}^2$$

# Fuzzy K-means algorithm

**Name:** **Fuzzy K-means clustering (stochastic optimization)**

**Input:** Instances: $\{\mathbf{x}_n; n = 1,2,\ldots,N\}$

$K$: Number of clusters

$\eta$: Learning parameter

**Output:** Clusters $\{\gamma_{nk;}\ k=1,\ldots,K;\ n = 1,\ldots,N\}$

Cluster means $\{\boldsymbol{\mu}_k;\ k = 1,\ldots,K\}$

**Code:** Initialize the centers of the clusters at random

E.g. $\boldsymbol{\mu}_k = \mathbf{x}_{n_k}; \quad k = 1,2,\ldots,K$

$\mathrm{n}_{\mathrm{epoch}} = 0$

repeat until convergence

$\mathrm{n}_{\mathrm{epoch}} = \mathrm{n}_{\mathrm{epoch}} + 1$

for $n = 1 : N$

$$d_{nk} = \left\| \mathbf{x}_n - \boldsymbol{\mu}_k \right\|, \quad k = 1,2,\ldots,K$$

Soft (fuzzy) cluster assignment

$$u_{nk} = \frac{d_{nk}^{-2}}{\sum_{j=1}^{K} d_{nj}^{-2}}; \quad k = 1,2,\ldots,K$$

$$\boldsymbol{\mu}_k = \boldsymbol{\mu}_k + \eta\, u_{nk}^2 \left( \mathbf{x}_n - \boldsymbol{\mu}_k \right)$$

end

end

# Mixtures of Gaussians

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k N(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

where $N(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\boldsymbol{\Sigma}_k|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_k)^T \cdot \boldsymbol{\Sigma}_k^{-1} \cdot (\mathbf{x}-\boldsymbol{\mu}_k)\right]$

- A mixture of Gaussians model may be obtained from of a more complex density with discrete **latent variables**

  Consider the random K-dim. latent variable $\mathbf{z} = \{z_k\}_{k=1}^{K}$
  (1-of-K encoding)

$$z_k \in \{0,1\}, \quad k = 1,2,\ldots,K; \qquad \sum_{k=1}^{K} z_k = 1$$

with probability distribution $p(\mathbf{z}) = \prod_{k=1}^{K} \pi_k^{z_k}; \quad \pi_k \equiv p(z_k = 1)$

$$0 \le \pi_k \le 1, \quad k = 1,2,\ldots,K; \qquad \sum_{k=1}^{K} \pi_k = 1$$

Assume: $p(\mathbf{x} \mid \mathbf{z}) = \prod_{k=1}^{K} N(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_k}$

$$\Rightarrow p(\mathbf{x}, \mathbf{z}) = \prod_{k=1}^{K} \left[\pi_k N(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\right]^{z_k}$$

$$\Rightarrow p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{x}, \mathbf{z}) = \sum_{k=1}^{K} \pi_k N(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

**Posterior probability of latent variable**

$$\Rightarrow \gamma_k \equiv p(z_k = 1 \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid z_k = 1)\, p(z_k = 1)}{p(\mathbf{x})} = \frac{\pi_k N(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j N(\mathbf{x}; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

# Maximum likelihood for mixtures of Gaussians

Assume we have the data $\{\mathbf{x}_n\}_{n=1}^N$

$$L\left(\{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^K; \{\mathbf{x}_n\}_{n=1}^N\right) = p\left(\{\mathbf{x}_n\}_{n=1}^N \middle| \{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^K\right) =$$

$$= \prod_{n=1}^N \sum_{k=1}^K \pi_k N\left(\mathbf{x}_n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)$$

$$LL\left(\{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^K; \{\mathbf{x}_n\}_{n=1}^N\right) = \log L\left(\{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^K; \{\mathbf{x}_n\}_{n=1}^N\right) =$$

$$= \sum_{n=1}^N \log\left(\sum_{k=1}^K \pi_k N\left(\mathbf{x}_n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)\right)$$

The parameters that **maximize the likelihood** are

$$\gamma_{nk} = \frac{\pi_k N\left(\mathbf{x}_n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right)}{\sum_{j=1}^K \pi_j N\left(\mathbf{x}; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j\right)} \qquad [\text{Posterior probability of } z_{nk}]$$

$$N_k = \sum_{n=1}^N \gamma_{nk}$$

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} \mathbf{x}_n$$

**Soft cluster assignment**

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_{nk} \left(\mathbf{x}_n - \boldsymbol{\mu}_k\right)\left(\mathbf{x}_n - \boldsymbol{\mu}_k\right)^T$$

# Mixtures of Gaussians: Expectation maximization

**Name:** **Expectation maximization**

**Input:** Instances: $\{\mathbf{x}_n; \quad n = 1,2,\ldots,N\}$

$K$ : Number of Gaussians in the mixture

**Output:** Clusters $\{\gamma_{nk;}\ k=1,\ldots,K;\ \ n=1,\ldots,N\}$

Cluster means $\{\boldsymbol{\mu}_k\ ;\ k=1,\ldots,K\}$

**Code:** Initialize the values of the parameters of the mixture

$$\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k; \quad k = 1,2,\ldots,K$$

Repeat

1. **Expectation step**

$$\text{Calculate } \gamma_{nk} = \frac{\pi_k N(\mathbf{x}_n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j N(\mathbf{x}; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)};$$

$$k = 1,2,\ldots,K; \quad n = 1,2,\ldots,N$$

2. **Maximization step**

$$N_k = \sum_{n=1}^{N} \gamma_{nk}; \qquad \pi_k = \frac{N_k}{N}$$

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n=1}^{N} \gamma_{nk} \mathbf{x}_n; \qquad \boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_{n=1}^{N} \gamma_{nk} (\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^T$$

$$k = 1,2,\ldots,K$$

until convergence (based on improvement of likelihood)

# Mixtures of Gaussians: Relation to K-means

Assume $\quad \boldsymbol{\Sigma}_k = \varepsilon \mathbf{I}, \quad \varepsilon \to 0^+$

$$N(\mathbf{x}_n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) = \frac{1}{(2\pi\varepsilon)^{D/2}} \exp\left\{ -\frac{\left\| \mathbf{x}_n - \boldsymbol{\mu}_k^* \right\|^2}{2\varepsilon} \right\}$$

$$\lim_{\varepsilon \to 0^+} \gamma_{nk} = \lim_{\varepsilon \to 0^+} \frac{\pi_k N(\mathbf{x}_n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j N(\mathbf{x}; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} =$$

$$= r_{nk} \equiv \begin{cases} 1 & \text{if } k = \arg \min_{j=1,2,\ldots,K} \left\| \mathbf{x}_n - \boldsymbol{\mu}_j \right\|^2 \\ 0 & \text{otherwise} \end{cases}$$

$$\lim_{\varepsilon \to 0^+} \log L\left( \{\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^{K}; \{\mathbf{x}_n\}_{n=1}^{N} \right)$$

$$= \lim_{\varepsilon \to 0^+} \sum_{n=1}^{N} \log\left( \sum_{k=1}^{K} \pi_k N(\mathbf{x}_n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right) \approx$$

$$\approx \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} \log\left( \pi_k N(\mathbf{x}_n; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right) \approx$$

$$\approx -\frac{1}{2\varepsilon} \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} \left\| \mathbf{x}_n - \boldsymbol{\mu}_k^* \right\|^2 + \text{ smaller terms } O(\varepsilon^0)$$

# Elliptical K-means

Estimate both **means** and **covariance matrices** based on **hard assignments** to components in the Gaussian mixture.

Excerpt from:
"Example-Based Learning for View-Based Human Face Detection" Kah-Kay Sung and Tomaso Poggio, IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 20, No. 1, January 1998]

The following is an outline of our **elliptical k-means clustering** procedure:

1) Obtain k (six in our case) initial pattern centers by performing vector quantization with Euclidean distances on the enlarged face database. Divide the data set into k partitions (clusters) by assigning each data sample to the nearest pattern center in Euclidean space.
2) Initialize the covariance matrices of all k clusters to be the identity matrix.
3) Recompute pattern centers to be the centroids of the current data partitions.
4) Using the current set of k pattern centers and their cluster covariance matrices, recompute data partitions by reassigning each data sample to the nearest pattern center in normalized Mahalanobis distance space. If the data partitions remain unchanged or if the maximum number of inner-loop (i.e., Steps 3 and 4) iterations has been exceeded, proceed to Step 5. Otherwise, return to Step 3.
5) Recompute the covariance matrices of all k clusters from their data partitions.
6) Using the current set of k pattern centers and their cluster covariance matrices, recompute data partitions by reassigning each data sample to the nearest pattern center in normalized Mahalanobis distance space. If the data partitions remain unchanged or if the maximum number of outer-loop (i.e., Steps 3 to 6) iterations has been exceeded, proceed to Step 7. Otherwise, return to Step 3.
7) Return the current set of k pattern centers and their cluster covariance matrices.

# Elliptical K-means for face detection

40 IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, VOL. 20, NO. 1, JANUARY 1998

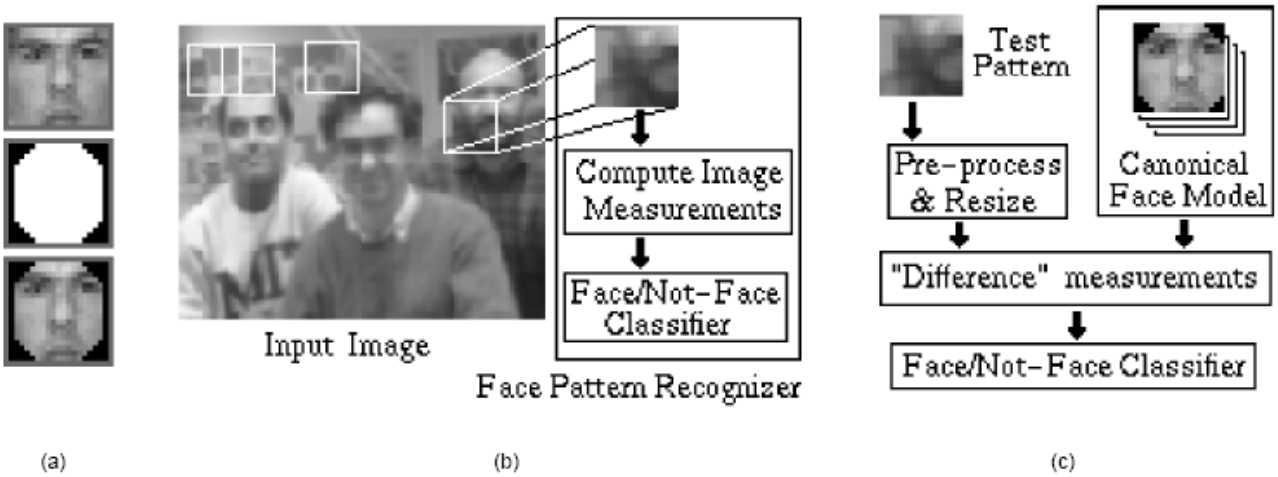



Fig. 1. Overview of our face-detection system. (a) A "canonical" face pattern, a 19 × 19 mask for eliminating near-boundary pixels of canonical face patterns, and the resulting "canonical" face pattern after applying the mask. (b) At each scale, the image is divided into many possibly overlapping windows. Each window pattern gets classified as either "a face" or "not a face," based on a set of local image measurements. (c) The key components of the *Face Pattern Recognizer* block in greater detail.

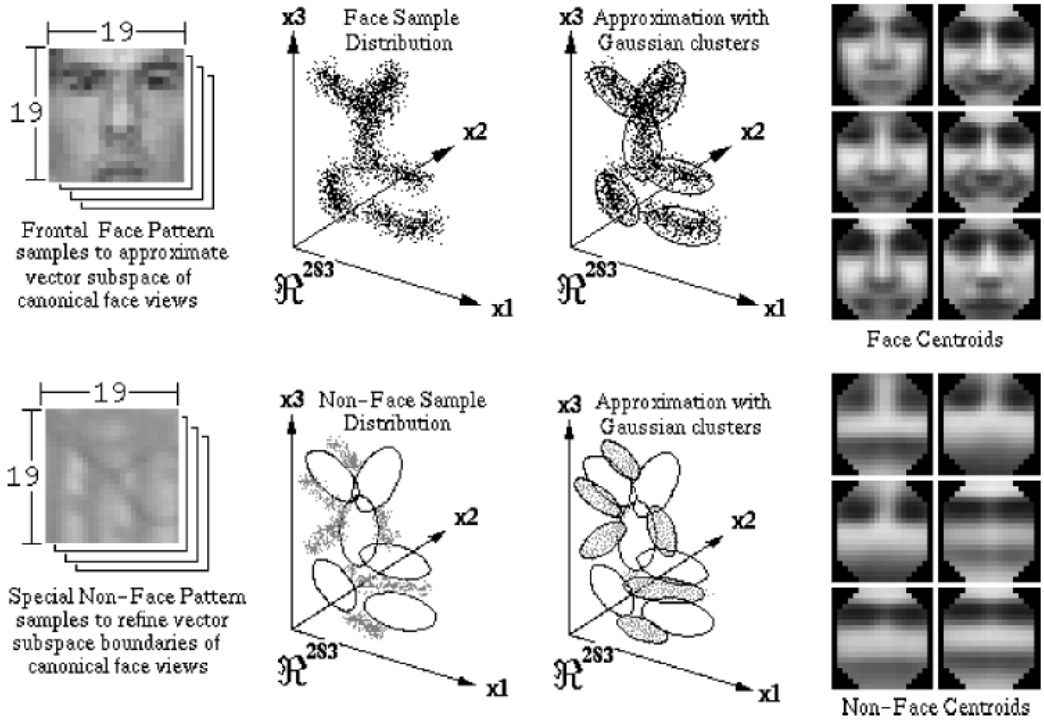
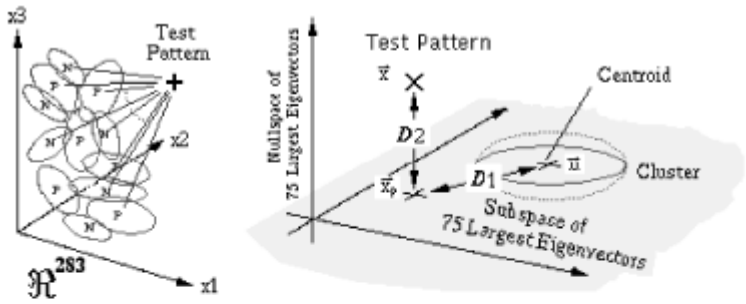


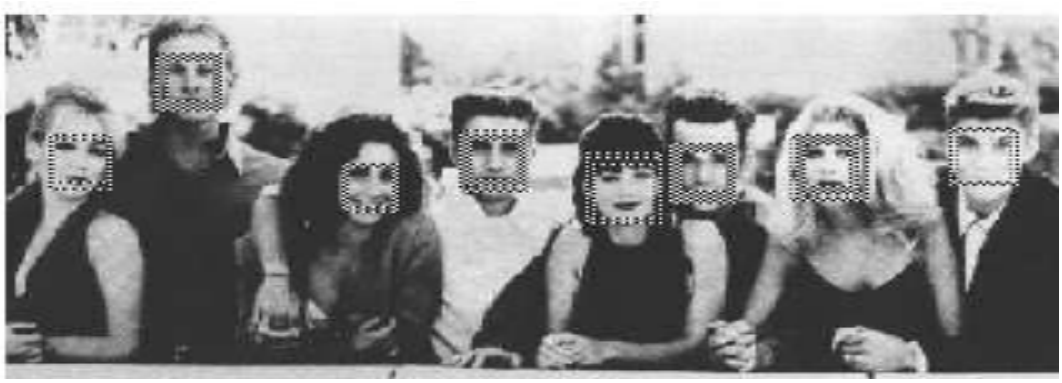

Fig. 2. Our distribution-based canonical face model. Top Row: We model an empirical distribution of face patterns using six multi-dimensional Gaussian clusters, whose centers are as shown on the right. Bottom Row: We also model a carefully chosen sample of nonface patterns using six multidimensional Gaussian clusters to help localize the boundaries of the face distribution. The final model consists of six Gaussian "face" clusters and six "nonface" clusters.

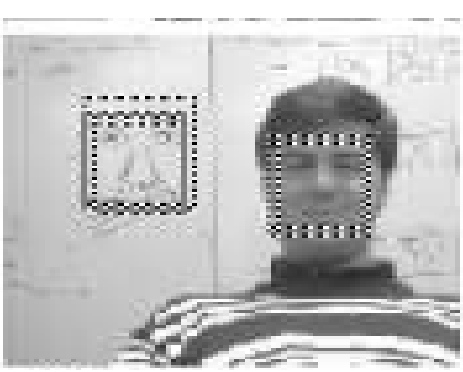
# Elliptical K-means for face detection




(a) (b)

Fig. 4. Matching a test pattern against our distribution-based model. (a) Each set of measurements is a vector of 12 distances between the test pattern and the model's 12 cluster centroids. (b) Each distance measurement between the test pattern and a cluster centroid is a two-value distance metric. $\mathcal{D}_1$ is a Mahalanobis distance between the test pattern's projection and the cluster centroid in a subspace spanned by the cluster's 75 largest eigenvectors. $\mathcal{D}_2$ is the Euclidean distance between the test pattern and its projection in the subspace.





# Kohonen SOM
# Self-organizing map

**Name:** **topological clustering (Kohonen SOM map)**

**Input:** Instances: $\{\mathbf{x}_n; n = 1,2,\ldots,N\}$

$K$: Number of clusters

$\eta$: Learning parameter

$\alpha$: Annealing hyperparameter

R: number of neighbors

$\text{neighborhood}(k;R)$: neighborhood operator for cluster $k$

**Output:** Cluster weights $\{\mathbf{w}_k;\ k = 1,\ldots,K\}$

**Code:** Initialize weights of clusters at random $\mathbf{w}_k; k = 1,2,\ldots,K$

$\mathrm{n}_{\text{epoch}} = 0$

repeat until convergence

$\mathrm{n}_{\text{epoch}} = \mathrm{n}_{\text{epoch}} + 1$

for $n = 1:N$

$$j = \min_{k=1,2,\ldots,K} \|\mathbf{x}_n - \mathbf{w}_k\|$$

$$\mathbf{w}_k = \mathbf{w}_k + \eta(\mathbf{x}_n - \mathbf{w}_k);\quad k \in \text{Neighborhood}(j;R)$$

end

$\eta = \alpha\eta$

Learning rate can be updated with other schedules (e.g. linear, instead of geometric)

Reduce $R$ at specified times

end